Sales Performance & Customer Insights Dashboard

Transforming Data into Actionable Insights 📈

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Role:

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Tools & Technologies:

Power BI | Microsoft Excel | DAX|SQL

Project Description:

This report presents an interactive Power BI dashboard developed to analyze sales performance, profitability, and customer behavior. The dashboard provides insights into trends, regional performance, and key business metrics to support data-driven decision-making.

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EXECUTIVE SUMMARY

This report presents a comprehensive analysis of sales performance and customer insights using an interactive Power BI dashboard. The objective of this project is to evaluate business performance across multiple dimensions, including sales trends, regional contributions, product categories, and customer behavior.

The dataset used for this analysis was sourced from an Excel file containing transactional sales data. The data was cleaned and prepared by removing duplicates, handling missing values, and standardizing inconsistent entries. A dedicated date table was created to enable accurate time-based analysis and trend evaluation.

The dashboard highlights key performance indicators such as total sales, total profit, total orders, and profit margin. Additionally, a month-over-month growth metric was developed using DAX to track business performance over time. Visualizations such as line charts, bar charts, and donut charts were used to represent trends and distributions effectively.

Key insights derived from the dashboard indicate that sales have shown an overall upward trend with noticeable fluctuations during certain periods. The West region emerged as the highest contributor to revenue, while the Technology category led in overall sales performance. It was also observed that some customers generated high revenue but contributed relatively lower profit margins.

This dashboard provides valuable insights that can assist stakeholders in making informed decisions. It enables businesses to monitor performance, identify growth opportunities, and optimize strategies for improved profitability.

Overall, the project demonstrates the effective use of Power BI, data modeling, and DAX calculations to transform raw data into meaningful business insights.

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OBJECTIVE

The objective of this project is to analyze sales performance and customer behavior using an interactive Power BI dashboard. The goal is to identify trends, evaluate key metrics, and provide actionable insights to support business decision-making.

SCOPE OF THE PROJECT

This project focuses on:

- Analyzing historical sales data
- Evaluating regional and category-wise performance
- Identifying high-value customers
- Tracking monthly sales trends
- Measuring profitability and growth

The scope is limited to the available dataset and does not include external market factors.

DATASET DESCRIPTION

The dataset used in this project is a sales dataset stored in Excel format. It contains transactional records that provide insights into business performance.

Key Columns:

• Order Date

• Sales

• Profit

• Quantity

• Category and Sub-Category

• Region

• Customer Name

The dataset enables analysis across time, geography, and customer segments.

DATA CLEANING & PREPARATION

The dataset was prepared using Microsoft Excel and Power BI to ensure accuracy, consistency, and usability for analysis.

Data Cleaning Steps:

• Removed duplicate records to avoid redundancy
• Handled missing or blank values where applicable
• Standardized inconsistent text values (e.g., region names)
• Converted the Order Date column into proper date format
• Ensured correct data types for all columns

Excel-Based Analysis (Preliminary Analysis):

Before importing the dataset into Power BI, initial analysis was performed using Excel to understand the data structure and identify patterns.

• Created Pivot Tables to summarize:

- Total Sales by Region
- Sales by Category
- Monthly Sales Trends
- Top 10 Customers
- Top 5 Products

• Used Pivot Charts to visualize:

- Sales distribution across regions
- Category-wise performance
- Time-based trends

• Identified:

- High-performing regions and categories
- Initial sales patterns and outliers
- Data inconsistencies that required cleaning

This preliminary analysis helped in better understanding the dataset and guided the design of the Power BI dashboard.

DATA MODELING

A structured data model was created in Power BI to support efficient analysis.

• Created relationship between Date Table and Sales Table
• Developed calculated columns:

❖ Month-Year
❖ Month Number

• Created key measures using DAX:

❖ Total Sales
❖ Total Profit
❖ Total Orders
❖ Profit Margin
❖ Sales Growth %

This model ensures accurate calculations and dynamic filtering.

DAX Measures Created:

The following measures were developed using DAX to support business analysis:

1. Total Sales
Calculates total revenue generated:

```
[Total Sales = SUM(Orders[Sales])]
```

2. Total Profit
Calculates overall profit:

```
[ Total Profit = SUM(Orders[Profit])]
```

3. Total Orders
Counts number of orders:

```
[ Total Orders = DISTINCTCOUNT(Orders[Order ID])]
```

4. Profit Margin
Calculates profitability ratio:

```
[ Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)]
```

5. Sales Last Month
Calculates previous month sales:

```
[Sales Last Month = CALCULATE([Total Sales],
DATEADD('Date Table'[Date], -1, MONTH))]
```

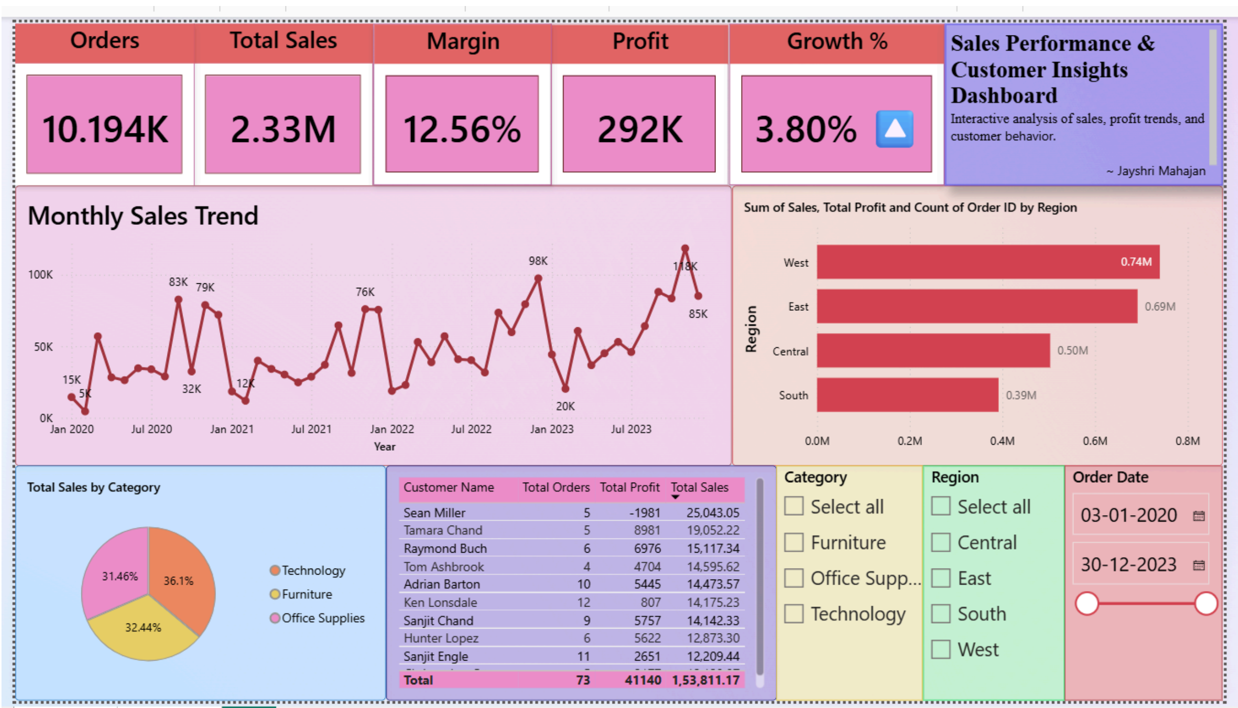
6. Sales Growth %

```
[Sales Growth Display =
FORMAT([Sales Growth %], "0.00%") & " " &
IF([Sales Growth %] > 0, "🔼", "🔽")]
```

These measures enable dynamic calculations and allow users to analyze performance trends effectively within the dashboard.

DASHBOARD OVERVIEW

The Power BI dashboard provides an interactive and visual representation of sales performance.



The dashboard includes:

- KPI cards for key metrics
- Monthly sales trend analysis
- Regional performance comparison
- Category distribution
- Customer insights

It allows users to explore data dynamically using filters.

Key Insights & Findings

1. **Overall Business Performance :** The business generated total sales of 2.33M across 10.19K orders, resulting in a total profit of 292K with an overall profit margin of 12.56%. This indicates healthy revenue generation with stable profitability.
2. **Positive Business Growth :** The company achieved a growth rate of 3.80%, showing steady business expansion over the analyzed period.
3. **Regional Performance Analysis :** The **West** region contributed the highest sales **(0.74M)**, followed by the **East** region **(0.69M)**.The **South** region generated the lowest sales **(0.39M)**, indicating potential opportunities for market expansion or strategy improvement.
4. **Monthly Sales Trends :** Sales performance showed fluctuations across months, with strong peaks in late periods**.** The highest monthly sales reached approximately **148K**, indicating seasonal demand or successful promotional campaigns. Some months showed lower performance, suggesting possible off-season trends or operational challenges.
5. **Category Contribution :** Among product categories:
    - Technology contributed the highest share **(36.1%)** of total sales.
    - Furniture contributed **32.44%**
    - Office Supplies contributed **31.46%**

   This suggests Technology products are the primary revenue drivers.

6. **Customer Performance :** A small group of customers contributed significantly to total sales and profitability, showing the importance of customer retention and relationship management.Some customers generated high sales but low profit, indicating discount or pricing impact.
7. **Profitability Opportunities :** Despite strong sales, profit performance varies across customers and regions.This suggests opportunities to optimize pricing strategy, discount policies, and product mix to improve margins.

Final Business Recommendation

Based on the analysis:

- Focus on expanding sales in the South region
- Strengthen marketing for high-performing Technology products
- Review discount strategies for low-profit customers
- Build retention strategies for top-value customers

BUSINESS IMPACT

This dashboard supports business decision-making by:

- Providing visibility into sales performance
- Identifying profitable regions and products
- Helping optimize customer targeting strategies
- Enabling trend analysis for forecasting

It enhances the ability to make data-driven decisions.

TOOLS & TECHNOLOGIES

- Power BI – Data visualization and dashboard creation
- Microsoft Excel – Data cleaning and preprocessing
- DAX – Data modeling and calculations
- SQL (MySQL) – Data storage, querying, and data import using structured queries

CONCLUSION

The project successfully demonstrates the use of Power BI to analyze sales data and generate meaningful insights. The dashboard provides a clear understanding of business performance and highlights key trends, enabling better strategic decisions.

This project showcases strong skills in data analysis, visualization, and business intelligence.